

AWS & GCP

Cloud Platforms Overview

1. Introduction

Amazon Web Services (AWS) and Google Cloud Platform (GCP) are leading public cloud platforms. They provide on-demand computing, storage, networking, databases, and managed services over the internet using a pay-as-you-go model. Cloud platforms reduce the need to purchase and maintain physical infrastructure, while allowing teams to provision resources quickly and scale capacity according to workload demand.

AWS, launched by Amazon in 2006, is widely recognized for its extensive service catalog, mature ecosystem, global infrastructure, and strong enterprise adoption. It is particularly suitable for organizations that need a broad range of infrastructure and managed services with substantial configuration flexibility.

GCP, launched by Google in 2008, is particularly strong in data analytics, machine learning and AI, and container orchestration. Google created Kubernetes, and GKE reflects that deep Kubernetes heritage. GCP is also well known for BigQuery and for its developer-focused cloud experience.

Both platforms use a shared responsibility model for security, provide geographically distributed regions, offer identity and access management, and support hybrid and multi-cloud architectures. The best choice depends on workload requirements, existing technology, cost considerations, and the organization's operational goals.

2. AWS — Core Services

Amazon EC2

Compute Provides resizable virtual machines in the cloud. EC2 offers instance families optimized for compute, memory, storage, and GPU workloads, with scaling and flexible purchasing options such as On-Demand, Reserved, and Spot Instances.

Typical use cases: Application servers, web workloads, scalable compute, lift-and-shift migrations.

Amazon S3

Object Storage Stores and retrieves objects such as files, media, backups, and datasets. S3 provides high durability, multiple storage classes, and integration with a wide range of AWS services.

Typical use cases: Backups, static assets, data lakes, media storage, archival.

AWS Lambda

Serverless Compute Runs code in response to events without requiring server management. It supports event-driven workloads and integrates with services such as S3, API Gateway, and DynamoDB.

Typical use cases: APIs, automation, event processing, lightweight microservices.

Amazon RDS

Managed Relational Database Provides managed relational databases including MySQL, PostgreSQL, MariaDB, Oracle, and SQL Server. It automates tasks such as backups, patching, and high-availability configurations.

Typical use cases: Transactional applications, structured data, managed SQL workloads.

Amazon VPC

Networking Creates an isolated virtual network in AWS with control over IP ranges, subnets, routing, gateways, security groups, and network ACLs.

Typical use cases: Secure multi-tier architectures, private networking, application isolation.

AWS IAM

Security & Access Controls authentication and authorization for AWS resources through users, groups, roles, and policies. IAM supports least-privilege access, MFA, and temporary credentials through roles.

Typical use cases: Identity management, workload permissions, secure AWS access.

Amazon DynamoDB

NoSQL Database Fully managed NoSQL database supporting key-value and document data models. It is designed for low-latency workloads and automatically handles scaling, partitioning, and replication.

Typical use cases: High-traffic applications, gaming, IoT, low-latency workloads.

CloudFormation / CodePipeline

IaC & CI/CD CloudFormation defines AWS infrastructure as code using JSON or YAML templates. CodePipeline automates build, test, and deployment stages. Together they support repeatable infrastructure and automated delivery.

Typical use cases: Infrastructure automation, CI/CD pipelines, repeatable deployments.

3. GCP — Core Services

Google Compute Engine (GCE)

Compute Provides scalable virtual machines running in Google's data centers. Custom machine types allow precise vCPU and memory allocation, and live migration can reduce disruption during host maintenance.

Typical use cases: General-purpose compute, application hosting, lift-and-shift migrations.

Google Cloud Storage (GCS)

Object Storage Stores unstructured data such as images, videos, backups, and datasets. Storage classes include Standard, Nearline, Coldline, and Archive for different access patterns.

Typical use cases: Data lakes, backups, media, content storage, analytics data.

Google Kubernetes Engine (GKE)

Managed Kubernetes Managed Kubernetes service for deploying, operating, and scaling containerized applications. GKE automates major cluster-management tasks and is closely aligned with Google's Kubernetes expertise.

Typical use cases: Microservices, cloud-native applications, container orchestration.

BigQuery

Data Warehouse Serverless, managed data warehouse designed for large-scale SQL analytics. It separates storage and compute and supports analytics across very large datasets.

Typical use cases: Business intelligence, large-scale analytics, data engineering, ML workflows.

Cloud Functions

Serverless Compute Runs event-driven code without requiring server management. Functions can respond to HTTP requests, file events, and messaging events.

Typical use cases: Automation, webhooks, event processing, lightweight APIs.

Cloud SQL

Managed Relational Database Managed relational database service supporting MySQL, PostgreSQL, and SQL Server. It handles routine operations such as backups, replication, patching, and failover.

Typical use cases: Transactional applications and managed relational workloads.

Cloud IAM

Security & Access Controls access to GCP resources using roles and policies across the organization, folder, project, and resource hierarchy.

Typical use cases: Identity management, least privilege, governance, secure access.

Pub/Sub & Cloud Build

Messaging & CI/CD Pub/Sub provides scalable asynchronous messaging for event-driven systems. Cloud Build automates build, test, and deployment workflows.

Typical use cases: Decoupled services, event-driven architectures, automated software delivery.

4. AWS vs GCP — Quick Comparison

Aspect	AWS	GCP
Market position	Largest cloud service ecosystem and strong enterprise adoption	Strong focus on data, AI/ML, analytics, and cloud-native workloads
Compute	Amazon EC2	Google Compute Engine
Object storage	Amazon S3	Google Cloud Storage
Serverless	AWS Lambda	Cloud Functions
Managed SQL database	Amazon RDS	Cloud SQL
NoSQL database	Amazon DynamoDB	Firestore / Bigtable
Managed Kubernetes	Amazon EKS	Google Kubernetes Engine (GKE)
Data warehouse	Amazon Redshift	BigQuery
Key strengths	Breadth of services, ecosystem, enterprise maturity	Data analytics, ML/AI, Kubernetes, data-centric workloads

5. Service Mapping for Cloud & DevOps

A useful way to understand the two platforms is to map services by architectural function rather than by product name. The following mapping highlights common building blocks used when designing cloud infrastructure and DevOps workflows.

Architecture Function	AWS	GCP
Compute	EC2	Compute Engine
Object Storage	S3	Cloud Storage
Serverless	Lambda	Cloud Functions
Relational Database	RDS	Cloud SQL
NoSQL Database	DynamoDB	Firestore / Bigtable
Kubernetes	EKS	GKE
Identity & Access	IAM	Cloud IAM
Infrastructure as Code	CloudFormation	Cloud Deployment Manager / IaC tooling
CI/CD	CodePipeline	Cloud Build
Messaging	SNS / SQS	Pub/Sub
Data Warehouse	Redshift	BigQuery

6. Key Takeaways

- AWS is a strong general-purpose choice when an organization needs a very broad ecosystem of infrastructure and managed services.

- GCP is particularly compelling for analytics, machine learning and AI, data-intensive workloads, and Kubernetes-native architectures.
- EC2 and Compute Engine provide comparable virtual-machine compute capabilities, while S3 and Cloud Storage serve the common object-storage role.
- AWS Lambda and Cloud Functions provide event-driven serverless execution without traditional server management.
- RDS and Cloud SQL simplify administration of relational databases, while DynamoDB and GCP's NoSQL offerings address different high-scale application patterns.
- EKS and GKE provide managed Kubernetes, reducing the operational burden of running Kubernetes clusters directly.
- IAM is a fundamental security layer on both platforms and should be designed around least privilege and controlled access.
- Infrastructure as Code and CI/CD services help make cloud environments repeatable, auditable, and easier to operate.

7. Conclusion

AWS and GCP both provide mature cloud platforms capable of supporting modern applications, DevOps workflows, and scalable infrastructure. AWS stands out for service breadth and enterprise maturity, while GCP has notable strengths in analytics, AI/ML, and Kubernetes. Understanding the equivalent services and the architectural role each service plays makes it easier to select the right platform for a specific workload rather than choosing based only on product names.